# NIT-T Hardware Foundry

## Ultimate End-to-End Initiative Guide

Prepared for the Hardware Domain Ideation Task, Technical Council Inductions 2026.

## 1. Executive Vision

NIT-T Hardware Foundry is a flagship beginner-to-builder hardware initiative for first-year students at NIT Trichy. The program is designed to take students from curiosity to confidence through live demonstrations, guided build studios, open-ended campus problem tracks, reusable hardware kits, mentor clinics, and a final public prototype expo.

The Year 1 target is 300 first-year students. The three-year growth goal is to scale the initiative into a campus-wide hardware pipeline serving 700 students per year by Year 3.

The event is intentionally ambitious. It should not feel like a normal workshop where students listen, copy a circuit, and leave. It should feel like entering a serious student-led invention studio. Students should leave with a working prototype, a failure log, a bill of materials, a team demo, and the belief that hardware is something they can build, repair, explain, improve, and lead.

The core message:

> **Hardware Foundry is not a talent filter. It is a confidence accelerator.**

## 2. One-Line Pitch

NIT-T Hardware Foundry is a 300-student prototype accelerator where first-years turn curiosity into working campus hardware solutions, then grow into the next generation of Technical Council builders and mentors.

## 3. Why This Initiative Matters

Hardware has a higher fear barrier than many other technical fields. In software, mistakes are often reversible by editing code. In hardware, a beginner faces messy wires, loose grounds, burnt LEDs, sensor noise, mechanical friction, bad tolerances, tool safety, and the fear of breaking something physical. This fear keeps many capable first-years away from electronics, robotics, embedded systems, CAD, mechanisms, and product prototyping.

The Foundry solves this by making the first hardware experience structured, safe, social, and meaningful.

Students do not simply learn "Arduino basics." They learn the full engineering loop:

- Observe a real problem.
- Ask better questions.
- Choose a small version of the problem.
- Build a working prototype.
- Test it.
- Fail safely.
- Debug it.
- Document it.
- Present it.
- Improve it.
- Teach the next student.

The visible output is a prototype. The deeper output is a campus culture of builders.

## 4. Research Anchors

The design of Hardware Foundry is based on three evidence-backed ideas.

### Active Learning

Active learning is better suited than passive lectures for beginner engineering. A large STEM meta-analysis summarized in PNAS/Wired found that active learning improved exam performance and reduced failure rates compared with traditional lecturing. Hardware Foundry converts that principle into a build-first format: short concept bursts, immediate practice, team debugging, and reflection.

### Project-Based Learning

Project-based learning is strongest when students work on open-ended challenges, make meaningful choices, receive feedback, revise their work, and present a public product. PBLWorks describes strong project design through elements such as a challenging problem, sustained inquiry, authenticity, student voice and choice, reflection, critique and revision, and public product. Hardware Foundry makes those elements explicit: students choose problem tracks, build prototypes, iterate through mentor clinics, and present in a public expo.

### CDIO Engineering Model

The CDIO framework describes engineering education through Conceive, Design, Implement, Operate. Hardware Foundry maps directly to this:

- Conceive: identify a campus problem and user need.
- Design: create a block diagram, CAD sketch, wiring plan, and BOM.
- Implement: build the electronics/mechanics prototype.
- Operate: test, demonstrate, document, and plan the next version.

### Design Thinking

The creativity model also borrows from design thinking: students observe a real context, define a sharper problem, ideate multiple possibilities, prototype a small version, and test it. This is why the Foundry does not give every team the same project. It gives them structured freedom: a safe boundary, a problem track, mentors, and enough room for original ideas.

## 5. NIT Trichy Context and Venue Logic

Venue choices must be approved by the relevant authorities. The plan below uses NIT-T facilities as permission-dependent candidates, not guaranteed bookings.

### Orion / Lecture Hall Complex

Use case:

- Opening ceremony.
- Big inspiration talk.
- Safety briefing.
- Final pitch orientation.

Why it fits:

- NIT-T's official website lists Lecture Hall Complex / Orion as a campus facility. A large lecture venue gives the launch seriousness and allows a 300-student opening without fragmenting the energy.

### Octagon / Computer Support Group

Use case:

- CAD and simulation bootcamps.
- Wokwi / circuit simulation sessions.
- Documentation and BOM clinics.
- Poster and report preparation.

Why it fits:

- NIT-T's Computer Support Group page describes Octagon and related computer facilities as central computing infrastructure with multiple labs, CAD/CAM references, and high-end computers. This makes it a strong candidate for simulation and digital design sessions.

## Central Workshop

Use case:

- Supervised fabrication exposure.
- Demonstration of tools and fabrication principles.
- Limited, permission-controlled mechanism and enclosure support.

Why it fits:

- NIT-T's Central Workshop page describes it as a common facility providing fabrication services for projects and prototype fabrication, including fitting/carpentry, machine shop, welding/foundry sections, and first-year fabrication training.

## SAC / Student Activity Spaces

Use case:

- Public demo expo.
- Poster display.
- Walkthrough showcase.
- Community gathering.

Why it fits:

- A public-facing student activity venue makes the final expo visible and aspirational.

## Department Classrooms and Seminar Rooms

Use case:

- Smaller studio cohorts.
- Build clinics.
- Mentor review tables.

Why it fits:

- The 300-student model should not put all teams into one build room. It should split them into cohorts and rotate through spaces.

## CEDI / Incubation Ecosystem

Use case:

- Optional demo-day jury.
- Entrepreneurship talk.
- Continuation track for promising prototypes.

Why it fits:

- CEDI is relevant as an entrepreneurship and incubation anchor. Hardware Foundry should not depend on CEDI for basic execution, but should connect high-potential prototypes to the incubation ecosystem after the event.

## 6. Scale Model

### Year 1 Target

- 300 students.
- 60 teams.
- 5 students per team.
- 10 studio clusters.
- 6 teams per studio cluster.
- 18-person core organizing team.

### Three-Year Growth Goal

| Year | Student Reach | Teams | Operating Model | Goal |
| --- | --- | --- | --- | --- |
| Year 1 | 300 | 60 | Core Foundry with 10 studio clusters | Build proof of concept and a hardware culture |
| Year 2 | 500 | 100 | Add Foundry Fellows and department-linked tracks | Convert the initiative into a repeatable campus pipeline |
| Year 3 | 700 | 140 | Multi-track Foundry Week plus semester continuation labs | Become the largest beginner hardware launchpad on campus |

### Why 300 Is Feasible

300 students is feasible only if the event is designed as a distributed studio system, not a single-room workshop.

The key operating choices:

- Teach core concepts in common launch sessions.
- Split hands-on build time into studio clusters.
- Use 60 teams of 5 rather than 100 teams of 3.
- Use project templates and modular kits.
- Run mentor clinics by topic, not by individual team.
- Use documentation checkpoints to prevent chaos.
- Use demo-day triage: all teams show posters; selected teams pitch on stage.

## 7. Core Organizing Team

The task asks for an initiative that can be conducted by fewer than 20 people. Hardware Foundry therefore uses a core organizing team of 18. This team designs and runs the initiative. Participants may take internal team roles, but the official organizing structure remains under 20.

| Role | Count | Responsibility |
| --- | --- | --- |
| Program Director | 1 | Overall ownership, permissions, final quality |
| Operations Lead | 1 | Calendar, venue schedule, attendance flow |
| Electronics Stream Leads | 3 | Arduino/ESP32, sensors, displays, debugging |
| Mechanics/CAD Stream Leads | 3 | CAD, structures, mechanisms, enclosure thinking |
| Systems Integration Leads | 2 | Power, wiring, reliability, test plans |
| Safety and Inventory Leads | 2 | Kits, component checkout, safety rules |

| Role | Count | Responsibility |
|---|---|---|
| Studio Managers | 3 | Coordinate studio clusters and mentor clinics |
| Documentation and Design Leads | 2 | Project cards, posters, reports, demo storytelling |
| Expo and Outreach Lead | 1 | Publicity, demo day, guest coordination |
| Total | 18 | Within the task limit |

# 8. Participant Model

## Who Should Join

The Foundry should prioritize first-year students who:

- Are curious about hardware but unsure where to start.
- Have not yet joined a technical project team.
- Are willing to work consistently.
- Are ready to document failures honestly.
- Want to build something useful, not just attend sessions.

The program should not select only experienced students. If only already-skilled students enter, the initiative fails its purpose.

## Cohort Composition

For 300 students:

- 60 percent absolute beginners.
- 25 percent students with light exposure to Arduino/CAD/school projects.
- 15 percent students with stronger prior experience who can anchor teams.

## Team Composition

Each team of 5 should include:

- 2 beginners with no hardware background.
- 1 student comfortable with coding or logic.
- 1 student interested in mechanics/CAD/design.
- 1 student with strong communication/documentation interest.

This keeps every team functional without making skill a gatekeeping mechanism.

# 9. Creativity Philosophy

This is one of the most important changes from a normal workshop.

Hardware Foundry should not tell every team to build the same thing. It should give students:

- A safe technical boundary.
- A problem field.
- A kit.
- Mentors.
- Examples.
- A prototype deadline.

Then students should be allowed to let their ideas flow.

The organizing team should not force originality by making every idea complicated. It should nurture originality by asking better questions:

- What problem have you personally noticed?
- Who would use this?
- What is the smallest useful version?
- What can go wrong in real use?
- How can you make it cheaper?
- How can you make it easier to repair?
- What would make someone trust this device?
- Can another first-year rebuild it from your documentation?

The event should feel like support, not control. We are always there to help teams think, test, simplify, and recover when things fail.

## 10. Main Event Structure

Hardware Foundry has three layers:

### Layer 1 - Common Spark

All 300 students attend the opening session.

Purpose:

- Create excitement.
- Show live demos.
- Explain safety.
- Set the culture.
- Introduce tracks and teams.

### Layer 2 - Studio Clusters

Students split into 10 studio clusters.

Each studio cluster:

- 30 students.
- 6 teams.
- One scheduled build block at a time.
- Rotates through electronics, CAD, systems, and documentation clinics.

### Layer 3 - Public Expo

All teams submit a poster and prototype table. Selected teams pitch on stage.

This prevents demo day from becoming unmanageable while still letting every team be visible.

## 11. Timeline

### Phase 0 - Preparation, 3 Weeks

This happens before students begin.

Week -3:

- Finalize core team.

- Confirm permission request list.
- Identify candidate venues: Orion/LHC, Octagon, department rooms, Central Workshop, SAC.
- Freeze student target at 300.
- Define project tracks.
- Prepare safety policy.

Week -2:

- Launch publicity.
- Open registrations.
- Train organizers.
- Prepare sample demos.
- Prepare kit inventory.
- Prepare project handbook.

Week -1:

- Close registrations.
- Select participants.
- Create teams and studio clusters.
- Release pre-reading.
- Label kits.
- Dry run opening session.

## Phase 1 - Spark Week

Goal:

- Make hardware feel exciting and reachable.

Sessions:

- Opening ceremony and demo night.
- Safety and build culture.
- First circuit lab: LED, resistor, button, serial monitor.
- First sensor lab: reading the physical world.

Deliverable:

- Every participant makes a small circuit work individually or in pairs.

## Phase 2 - Skill Week

Goal:

- Give students enough tools to invent safely.

Sessions:

- Sensors and thresholds.
- Actuators and outputs.
- Displays and feedback.
- CAD and enclosure basics.
- Debugging clinic.

Deliverable:

- Every team submits a problem statement, system map, and first prototype sketch.

## Phase 3 - Idea Flow Week

Goal:

- Turn student curiosity into project direction.

Activities:

- Problem walk: students list observed campus/hostel/lab issues.
- Idea storm: 30 ideas in 30 minutes.
- Constraint filter: cost, safety, time, components.
- Mentor review: choose the smallest useful version.
- Team pitch: 60-second concept.

Deliverable:

- Each team submits a Foundry Proposal Card.

## Phase 4 - Build Sprint Week

Goal:

- Build the minimum working prototype.

Activities:

- Studio build blocks.
- Electronics clinic.
- CAD/mechanics clinic.
- Systems reliability clinic.
- Documentation clinic.

Deliverable:

- Minimum working prototype with at least one input, one processing decision, and one useful output.

## Phase 5 - Reliability and Story Week

Goal:

- Make projects demo-ready and understandable.

Activities:

- Failure mode review.
- Wiring cleanup.
- Enclosure/mounting improvement.
- BOM and poster creation.
- Demo rehearsal.

Deliverable:

- Prototype, poster, BOM, failure log, 90-second demo script.

## Phase 6 - Foundry Expo

Goal:

- Make hardware visible to the campus.

Format:

- 60 prototype tables.
- 60 project posters.
- Walkthrough judging.
- 12 shortlisted stage pitches.
- Awards for technical and learning outcomes.

Deliverable:

- Public demo and archived project repository.

### Phase 7 - Continuation

Goal:

- Prevent the event from ending as a one-time memory.

Actions:

- Select 30 Foundry Fellows.
- Publish project archive.
- Identify 10 prototypes for continuation.
- Invite top teams into Technical Council/CDI work.
- Train Fellows to mentor Year 2.

## 12. Project Tracks

Students should choose their own ideas within safe tracks. These tracks provide direction without killing creativity.

### Track A - Hostel Intelligence

Possible ideas:

- Water overflow alert.
- Mess queue counter.
- Common room occupancy indicator.
- Noise-level awareness display.
- Smart study-room availability board.

### Track B - Lab Helper Devices

Possible ideas:

- Cable continuity tester.
- Component sorting assistant.
- Mini lab equipment status display.
- Tool checkout reminder device.
- Bench safety checklist device.

### Track C - Campus Sustainability

Possible ideas:

- Dustbin fill indicator.
- Tap leakage alert.

- Energy reminder for rooms.
- Plant moisture monitor.
- Solar phone-charging status display.

### Track D - Accessibility and Inclusion

Possible ideas:

- Door-open indicator.
- Vibration alert device.
- Desk-edge obstacle alert.
- Low-cost tactile navigation marker prototype.
- Visual-to-sound alert prototype.

### Track E - Micro-Automation

Possible ideas:

- Servo-based sorting gate.
- Mini conveyor counter.
- Smart locker proof-of-concept.
- Timed dispenser prototype.
- Tabletop automated queue token indicator.

### Track F - Open Wildcard

For teams with strong ideas outside the above tracks.

Condition:

- Must pass safety, budget, feasibility, and documentation checks.

## 13. Flagship Project Family

The broad project family should be called:

### Campus Utility Sentinel

Each team builds one small hardware system that senses a real condition, makes a decision, and communicates a useful output.

Minimum requirements:

- One physical input.
- One processing decision.
- One local output.
- One safe enclosure or mounting plan.
- One test result.
- One failure analysis.

This structure keeps projects comparable without forcing every team to build the same device.

## 14. Teaching Model

The teaching model is:

**Micro-teach. Build. Debug. Reflect. Improve. Teach back.**

Each session should follow this rhythm:

| Segment | Duration | Purpose |
| --- | --- | --- |
| Problem story | 10 min | Make the concept meaningful |
| Concept burst | 15 min | Teach only what is needed now |
| Guided mini-build | 25 min | Everyone touches hardware |
| Team build | 50 min | Apply the idea creatively |
| Debug clinic | 20 min | Normalize failure and recovery |
| Reflection log | 10 min | Capture learning before it disappears |

## 15. What We Want Students to Feel

Students should feel:

- "Hardware is not magic."
- "I can ask basic questions."
- "Failure is data."
- "My idea is worth exploring."
- "I can build a small version first."
- "I can contribute even if I am not the best coder."
- "I can teach what I learned."

This emotional design matters. Confidence is the foundation of future technical leadership.

## 16. What Students Leave With

Every student should leave with:

- A working team prototype.
- A personal hardware starter checklist.
- A project poster.
- A bill of materials.
- A failure log.
- A 90-second demo experience.
- A map of what to learn next.
- A clear invitation to keep building.

Every team should leave with:

- A rebuildable project folder.
- Wiring diagram.
- Code.
- CAD/sketch/enclosure notes.
- Photos or demo video.
- Next-version plan.

# 17. Leadership Pipeline

The three-year 700-student goal depends on a leadership pipeline.

## Foundry Fellows

At the end of Year 1:

- Select 30 students from the 300.
- Selection is based on growth, documentation, teamwork, and ability to teach.
- Fellows receive extra training.
- Fellows help run Year 2.

## Year 2 Mentor Engine

Year 2:

- 500 participants.
- 50-70 Foundry Fellows and senior builders as support.
- More project tracks.
- More reusable kits.

## Year 3 Campus Hardware Network

Year 3:

- 700 participants.
- Department-linked tracks.
- Stronger sponsorship.
- More public expo presence.
- Alumni and industry demo feedback.

The final goal is not only attendance. The goal is a self-renewing hardware ecosystem.

# 18. Growth Roadmap to 700

| Metric | Year 1 | Year 2 | Year 3 |
|---|---|---|---|
| Students reached | 300 | 500 | 700 |
| Teams | 60 | 100 | 140 |
| Studio clusters | 10 | 16 | 20+ |
| Foundry Fellows | 30 | 75 | 125 |
| Project tracks | 6 | 8 | 10 |
| Public prototypes | 60 | 100 | 140 |
| Continuation projects | 10 | 20 | 35 |

## Year 1 Goal

Prove the model.

## Year 2 Goal

Make the model repeatable.

## Year 3 Goal

Make the model institutional.

## 19. Poster and Publicity Plan

### Poster 1 - Teaser

Headline:

| |
|---|
| **What if your first month at NIT-T ended with a working hardware prototype?** |

Purpose:

- Spark curiosity.

Visual:

- Breadboard, sensor, CAD sketch, glowing display, and workbench.

### Poster 2 - Myth Breaker

Headline:

| |
|---|
| **Hardware is not only robots.** |

Body:

| |
|---|
| **It is every sensor, switch, mechanism, enclosure, and product that notices the world and responds.** |

Purpose:

- Make hardware feel broad and inclusive.

### Poster 3 - Recruitment

Headline:

| |
|---|
| **NIT-T Hardware Foundry: 300 first-years. 60 teams. One campus-scale prototype expo.** |

Purpose:

- Communicate ambition.

### Poster 4 - Creativity

Headline:

| |
|---|
| **Bring a problem. Leave with a prototype.** |

Purpose:

- Encourage students to bring their own observations.

### Poster 5 - Mentor Culture

Headline:

| |
|---|
| **Built something? Help someone build their first thing.** |

Purpose:

- Recruit patient senior mentors and future Foundry Fellows.

### Poster 6 - Expo

Headline:

| |
|---|
| **Come see what first-years built from scratch.** |

Purpose:

- Make the final showcase visible.

## 20. Curiosity Prompts

Use these prompts in registration, posters, and opening sessions:

- What campus problem do you wish someone measured?
- What daily inconvenience could be solved with one sensor?
- What would you build if failure was allowed?
- What machine do you wish existed in your hostel?
- What simple device would help a lab run better?
- What would make campus more accessible?
- What can you prototype in four weeks that someone can understand in 30 seconds?

## 21. Out-of-Box Thinking System

Creativity needs structure. Use these constraints to improve ideas:

- Make it cheaper.
- Make it repairable.
- Make it usable without a phone.
- Make it work without Wi-Fi.
- Make it safe for a beginner.
- Make it explainable to a non-engineer.
- Make it survive rough handling.
- Make it modular.
- Make it useful even when imperfect.

## 22. Failure Culture

Every team must maintain a failure log.

The log must include:

- What failed.
- What the team assumed.
- What test revealed the truth.
- How the team fixed or simplified it.
- What they would do differently next time.

Awards should include:

- Best Failure Log.
- Best Debugging Discipline.
- Best Recovery.

This turns failure from embarrassment into engineering evidence.

## 23. Budget Model

### Year 1 Lean Budget - INR 60,000

Works if components are reused and fabrication is minimal.

| Item | Cost |
| --- | --- |
| 60 basic team kits | INR 30,000 |
| Shared sensor/actuator pool | INR 12,000 |
| Wires, breadboards, resistors, LEDs, buttons | INR 6,000 |
| Posters, project cards, printing | INR 4,000 |
| Storage boxes and labels | INR 3,000 |
| Replacement/contingency | INR 5,000 |

### Year 1 Recommended Budget - INR 1,20,000

Best balance for a 300-student flagship version.

| Item | Cost |
| --- | --- |
| 60 reusable team kits | INR 48,000 |
| Sensor library | INR 18,000 |
| Actuator/display pool | INR 14,000 |
| CAD/mechanical prototyping materials | INR 10,000 |
| Multimeters and tool pool | INR 12,000 |
| Posters, certificates, demo boards | INR 8,000 |
| Storage/inventory system | INR 4,000 |
| Contingency | INR 6,000 |

### Year 1 Premium Budget - INR 2,00,000

Use if sponsorship is available.

Adds:

- More ESP32 boards.
- Better displays.
- Extra multimeters.
- Enclosure materials.
- Demo-day branding.
- Awards.
- Dedicated continuation fund for top prototypes.

## 24. Kit Strategy

Each team kit:

- Arduino-compatible board or ESP32.
- Breadboard.
- Jumper wires.
- LEDs.
- Resistors.
- Buttons.
- Buzzer.
- One display module or LCD allocation.

- Basic sensors.
- Small servo allocation.

Shared pool:

- Ultrasonic sensors.
- LDRs.
- Temperature/humidity sensors.
- Water-level sensors.
- Current sensors.
- Extra displays.
- Servos.
- Motor drivers for approved teams.
- Fasteners.
- Foam board/acrylic/cardboard.

## 25. Safety Policy

Absolute rules:

- No mains voltage.
- No unsupervised soldering.
- No drilling/cutting without approved supervision.
- No lithium battery charging experiments without qualified supervision.
- No workshop tools outside approved spaces.
- No team works alone after approved hours.
- No component hoarding.
- No unsafe demo on expo day.

Safe default scope:

- USB power.
- 5V/9V low-current electronics.
- Breadboards.
- Sensors.
- Displays.
- Small servos.
- Simulation.
- CAD.
- Foam board/cardboard/acrylic mockups.

## 26. Documentation System

Each team maintains a project folder:

- Problem statement.
- User/context observation.

- System block diagram.
- Wiring diagram.
- Code.
- CAD/sketch.
- Bill of materials.
- Test log.
- Failure log.
- Photos/video.
- Final poster.
- Next-version plan.

Documentation is not a final chore. It is a weekly checkpoint.

## 27. Evaluation Rubric

| Category | Weight | What Judges Look For |
| --- | --- | --- |
| Problem clarity | 15 | Real need, clear user, focused scope |
| Creativity | 15 | Original thinking, thoughtful use of constraints |
| Technical implementation | 20 | Working input, logic, output, safe wiring |
| Systems thinking | 15 | Power, mounting, reliability, user handling |
| Iteration and failure learning | 15 | Tests, mistakes, fixes, simplification |
| Documentation | 10 | BOM, code, diagrams, project folder |
| Demo communication | 10 | Clear explanation, honest pitch, next steps |

## 28. Awards

Awards should reward the culture we want.

- Best Campus Utility Prototype.
- Best Creative Leap.
- Best Beginner Growth.
- Best Failure Log.
- Best Low-Cost Engineering.
- Best Documentation.
- Best Team Teaching Moment.
- Best Future Mentor.
- Best Accessibility Idea.
- Best Sustainability Idea.

## 29. Failure Points and Countermeasures

| Failure Point | Why It Matters | Countermeasure |
| --- | --- | --- |
| 300 students overwhelm organizers | Chaos reduces learning | Split into 10 studio clusters and 60 teams |
| Too lecture-heavy | Students disengage | Use micro-teach/build/debug rhythm |
| Components run out | Teams stall | Use reusable kits and checkout system |

| Failure Point | Why It Matters | Countermeasure |
|---|---|---|
| Safety incident | Major risk | Low-voltage only and strict tool policy |
| Copy-paste projects | Weak learning | Require failure log and oral subsystem explanation |
| One student dominates | Others do not learn | Rotate team roles weekly |
| Ideas become too ambitious | Prototypes fail | Require smallest useful version by Week 3 |
| Venue conflict | Schedule breaks | Maintain backup classrooms and staggered sessions |
| Documentation poor | Projects cannot continue | Weekly documentation checkpoints |
| Expo too crowded | Judges miss good work | Poster walk for all; stage pitch for shortlisted teams |
| Dropouts | Teams shrink | Use 5-person teams and attendance checkpoints |
| Mentor burnout | Quality drops | Use clinic model, not one-on-one dependency |
| Budget rejection | Event blocked | Present lean, recommended, and premium budget tiers |

## 30. Team Roles

Each 5-person team rotates these roles:

| Role | Responsibility |
|---|---|
| Circuit Lead | Wiring, sensors, power, safety |
| Code Lead | Embedded logic, display, state machine |
| Build Lead | Mounting, enclosure, mechanism |
| Test Lead | Test plan, measurements, reliability |
| Story Lead | Documentation, poster, demo pitch |

No one keeps the same role the whole time. Everyone learns more than one skill.

## 31. Operating Calendar

### T-30 to T-21 Days

- Build core team.
- Confirm event concept.
- Identify venues.
- Draft budget.
- Prepare permissions.

### T-20 to T-14 Days

- Launch publicity.
- Open registration.
- Prepare kit list.
- Create project handbook.

### T-13 to T-7 Days

- Select participants.
- Form teams.
- Create studio clusters.
- Train organizers.

- Freeze safety policy.

### T-6 to T-1 Days

- Label kits.
- Test demo circuits.
- Dry-run opening.
- Publish participant instructions.

### Event Weeks

- Week 1: Spark and basics.
- Week 2: Skills and systems.
- Week 3: Idea flow and proposal.
- Week 4: Build sprint.
- Week 5: Reliability and story.
- Week 6: Expo and continuation.

## 32. Final Expo Design

Expo layout:

- 60 tables.
- 10 zones matching studio clusters.
- One poster per team.
- Prototype on table.
- QR code to project folder.
- Judges rotate by zone.

Stage format:

- 12 shortlisted teams.
- 90 seconds pitch.
- 60 seconds questions.

Audience:

- First-years.
- Senior students.
- Technical Council members.
- Faculty mentors.
- CEDI/startup ecosystem representatives if available.
- Club/project team representatives.

## 33. Three-Year Institutionalization

### Year 1: Create the Foundry

Output:

- 300 participants.

- 60 project folders.
- 30 Foundry Fellows.
- 10 continuation prototypes.

### Year 2: Expand the Foundry

Output:

- 500 participants.
- 100 project folders.
- 75 Fellows.
- Department-linked project tracks.
- Better kit inventory.

### Year 3: Make It a Campus System

Output:

- 700 participants.
- 140 project teams.
- A standing Foundry Fellow network.
- Annual expo recognized across campus.
- Stronger sponsorship and incubation pathways.
- A project archive that becomes the beginner hardware knowledge base.

## 34. Why It Fits Technical Council

Hardware Foundry fits Technical Council because it:

- Makes technology accessible.
- Builds beginner confidence.
- Creates reusable technical resources.
- Encourages campus problem solving.
- Connects electronics, mechanics, CAD, software, design, and leadership.
- Produces visible prototypes.
- Builds future Technical Council contributors.
- Can start with fewer than 20 organizers.
- Can scale over three years.

## 35. Final Submission Summary

NIT-T Hardware Foundry is a 300-student beginner-to-builder prototype accelerator for first-year students. It uses permission-dependent campus venues such as lecture halls, computing facilities, workshop access, department rooms, and student activity spaces to run a distributed studio model. Students form 60 teams and build open-ended campus utility prototypes using reusable kits, active learning, mentor clinics, safety rules, failure logs, and public demo culture. The initiative is designed to scale to 700 students in three years by creating Foundry Fellows, reusable kits, project archives, and a repeatable studio-cluster operating model.

## 36. Sources and Research Notes

- NIT Trichy official homepage and facilities navigation: https://www.nitt.edu/
- NIT Trichy Central Workshop official page: https://www.nitt.edu/home/students/facilitiesnservices/centralworkshop/
- NIT Trichy Computer Support Group / Octagon official page: https://www.nitt.edu/home/students/facilitiesnservices/ComputerSupportGroup/
- NIT Trichy Lecture Hall Complex official page: https://www.nitt.edu/home/students/facilitiesnservices/lhc/
- CDIO official framework: https://www.cdio.org/
- CDIO about page, including Conceive-Design-Implement-Operate framing: https://www.cdio.org/about
- Active learning STEM evidence summary: https://www.wired.com/2014/05/empzeal-active-learning/
- Active learning meta-analysis DOI: https://www.pnas.org/doi/10.1073/pnas.1319030111
- PBLWorks Gold Standard Project Design: https://www.pblworks.org/what-is-pbl/gold-standard-project-design
- Stanford d.school Design Thinking Bootleg: https://dschool.stanford.edu/tools/design-thinking-bootleg